

DSA 8420 Spring 2025

Project 1

Due March 11, 2025

This project consists of two problems. In the first problem, we solve and analyze the agricultural problem presented in Homework 2. In the second problem, we examine a linear system with conflicting equations and compute a measure of its inconsistency.

1 The Agricultural Problem

Problem description. A farmer owns a farm that produces corn, soybean, and oats. There are 12 acres of land available for cultivation. Each crop that is planted has certain requirements for labor and capital. These data along with the net profit figure are given in the accompanying table.

	<i>Labor (hr)</i>	<i>Capital (\$)</i>	<i>Net profit (\$)</i>
Corn (per acre)	6	36	40
Soybeans (per acre)	6	24	30
Oats (per acre)	2	18	20

The farmer has \$360 available for capital and knows that there are 48 hr available for working these crops. How much of each crop should be planted to maximize profit?

Modeling. Let x_1, x_2, x_3 represent the number of acres we use to plant corn, soybeans, and oats, respectively. The problem can be modeled as a

linear program:

$$\begin{array}{ll}
 \max & 40x_1 + 30x_2 + 20x_3 & (\text{profit}) \\
 s.t. & x_1 + x_2 + x_3 \leq 12 & (\text{land}) \\
 & 6x_1 + 6x_2 + 2x_3 \leq 48 & (\text{labor}) \\
 & 36x_1 + 24x_2 + 18x_3 \leq 360 & (\text{capital}) \\
 & x_1, x_2, x_3 \geq 0 &
 \end{array}$$

Tasks.

- (18 points) With the accompanying data file ‘`agriculture.dat`’, develop an **abstract model** using Pyomo to solve the problem. Your code should take data from ‘`agriculture.dat`’ and print to screen the optimal solution and its corresponding optimal value. (Hint: You can use the Jupyter Notebook file in Homework 7 as a template.)
- Adjust the parameters (`avail_land`, `avail_labor`, `avail_capital`) in ‘`agriculture.dat`’ slightly away from the current value of (12, 48, 360) and re-run the code to answer the following three (independent) sets of questions.

Set 1:

- (2 points) Modify the value of ‘`avail_land`’ to complete the second row of the table below.

$$\text{avail_labor} = 48, \quad \text{avail_capital} = 360$$

avail_land (acre)	9	10	11	12	13	14	15
maximum profit (\$)							

- (2 points) Around the current value of the parameters, what is the rate of change of the profit with respect to the acres of land?
- (2 points) If one acre of land is available for rent during the period of cultivation, what is the highest price the farmer would like to pay? (The price is called the *shadow price* of the land.)

Set 2:

- (2 points) Modify the value of ‘`avail_labor`’ to complete the second row of the table below.

$$\text{avail_land} = 12, \quad \text{avail_capital} = 360$$

avail_labor (hr)	45	46	47	48	49	50	51
maximum profit (\$)							

- (e) (2 points) Around the current value of the parameters, what is the rate of change of the profit with respect to the hours of labor?
- (f) (2 points) If the farmer wants to hire a helper with the same efficiency, how much at most would the farmer pay the helper for an hour of work? (The rate is the shadow price of the labor.)

Set 3:

- (g) (2 points) Modify the value of ‘avail_capital’ to complete the second row of the table below.

$$\text{avail_land} = 12, \quad \text{avail_labor} = 48$$

avail_capital (\$)	330	340	350	360	370	380	390
maximum profit (\$)							

- (h) (2 points) Around the current value of the parameters, what is the rate of change of the profit with respect to the dollars of capital?
- (i) (2 points) If a bank offers to lend the farmer one dollar to expand his capital investment (hypothetically, don’t laugh), what is the highest acceptable interest rate for the farmer? (The rate is the shadow price of the capital.)

2 Inconsistency of A Linear System

Problem description. Consider the following system of linear equations:

$$2x_1 + 3x_2 = 6$$

$$2x_1 + 3x_2 = 8$$

$$x_1 + x_2 = 4$$

Clearly, this system is inconsistent – there is no pair (x_1, x_2) that simultaneously satisfies all three equations. Given any (x_1, x_2) , the absolute error for the first equation is defined as $|2x_1 + 3x_2 - 6|$. Similarly, the absolute error for the second and third equation is defined as $|2x_1 + 3x_2 - 8|$ and $|x_1 + x_2 - 4|$, respectively.

Modeling. We measure the inconsistency of the linear system by finding the minimum possible value of the maximum absolute error across all three equations. In mathematical terms, we seek to solve the following optimization problem:

$$\min_{x_1, x_2} \max\{|2x_1 + 3x_2 - 6|, |2x_1 + 3x_2 - 8|, |x_1 + x_2 - 4|\}$$

Tasks.

3. (6 points) Reformulate the above optimization problem as a linear program.
4. (18 points) Build a **concrete model** using Pyomo to solve the linear program. Your code should display the optimal solution and its corresponding optimal value.

Submission.

The submission consists of two parts:

- Part I: Submit your model code **to Canvas** as two Jupyter Notebooks—one for the abstract model for Task 1 and another for the concrete model for Task 4.
- Part II: Combine the remainder of your work into a PDF file and submit it **to Gradescope**. The PDF should include:
 - A screenshot of the final output of your code from Task 1. Be sure to highlight or separately list the optimal value and the optimal solution.
 - Tables and answers to the questions in Task 2.
 - The linear programming formulation from Task 3.
 - A screenshot of the final outputs of your code from Task 4. Again, highlight or separately list the optimal value and the optimal solution.